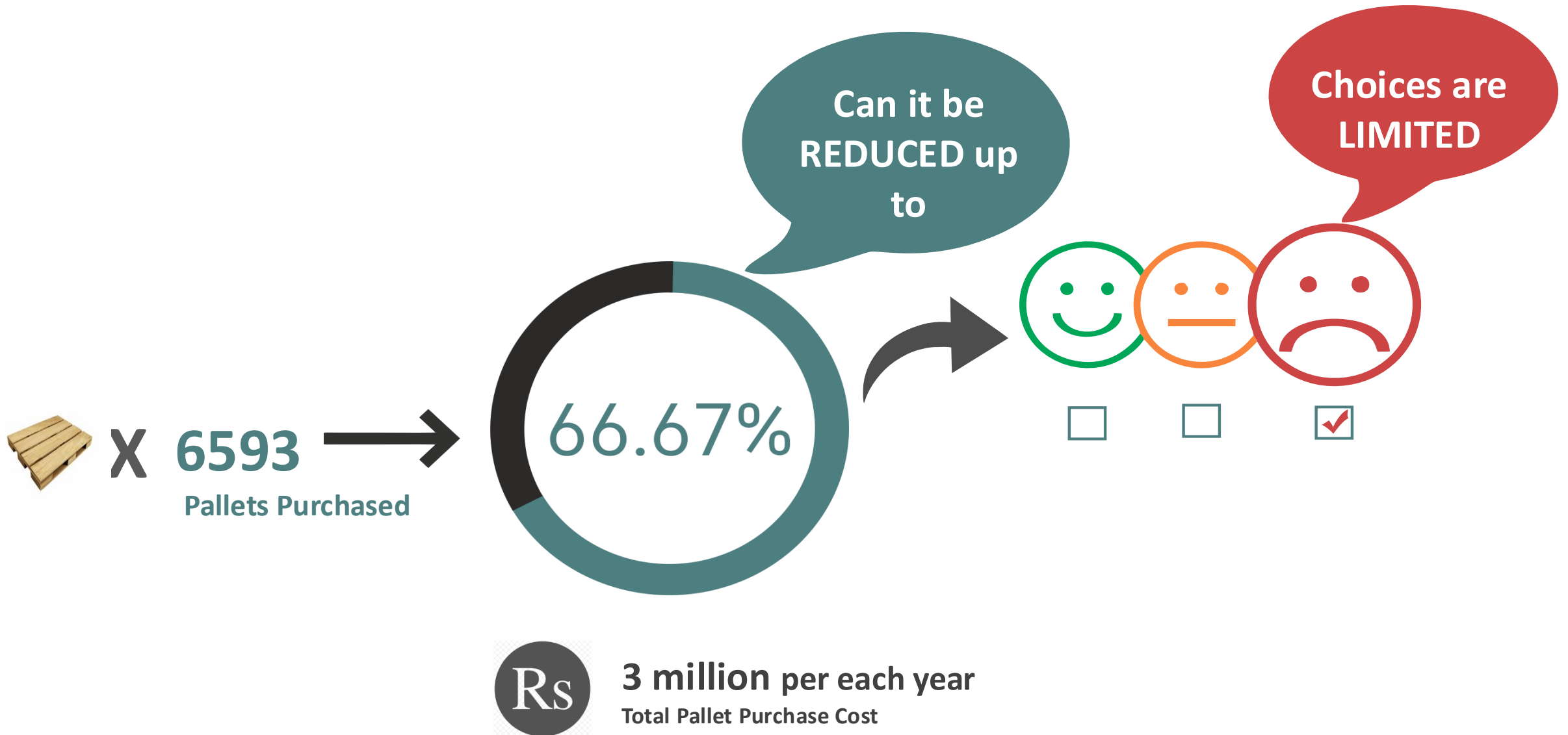




“Pallet Reverse Logistics Project”

Project OBJECTIVE – 2018/2019.



Project SCOPE.

Revolutionizing the traditional pallet purchasing practice and facilitating a

UNIQUE METHODOLOGY

Existing Practices?

Maximization of the Pallet Occupancy
Recycling & Reuse
Pallet Consumption Rate
Pallet Segregation Plan
Suitability of Existing Pallet Occupying
Damage Pallet Monitoring
Standardization of Pallet Purchasing

↑ Most Successful 😊

↓ Least Successful 😞



1. Cost saving through optimization.
2. Establishing control points.
3. Identify the pallet requirement and take preventive actions.
4. Keep the maximum supply chain visibility (smoothness) without any bottlenecks.
5. Segregating the pallet purchasing authority and make plant managers responsible.

Pallet Statistics in year 2019.



Damages

+



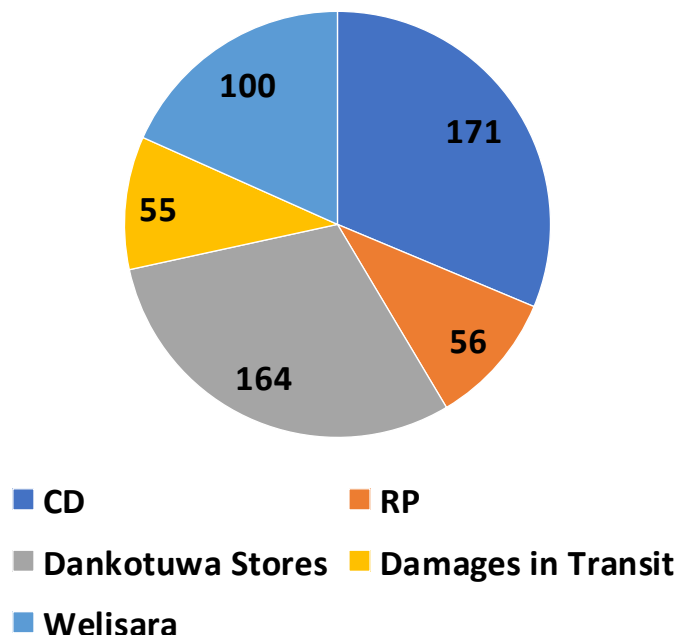
Disposals

+



Purchasing & Repair

Monthly Average = 545 Pallets

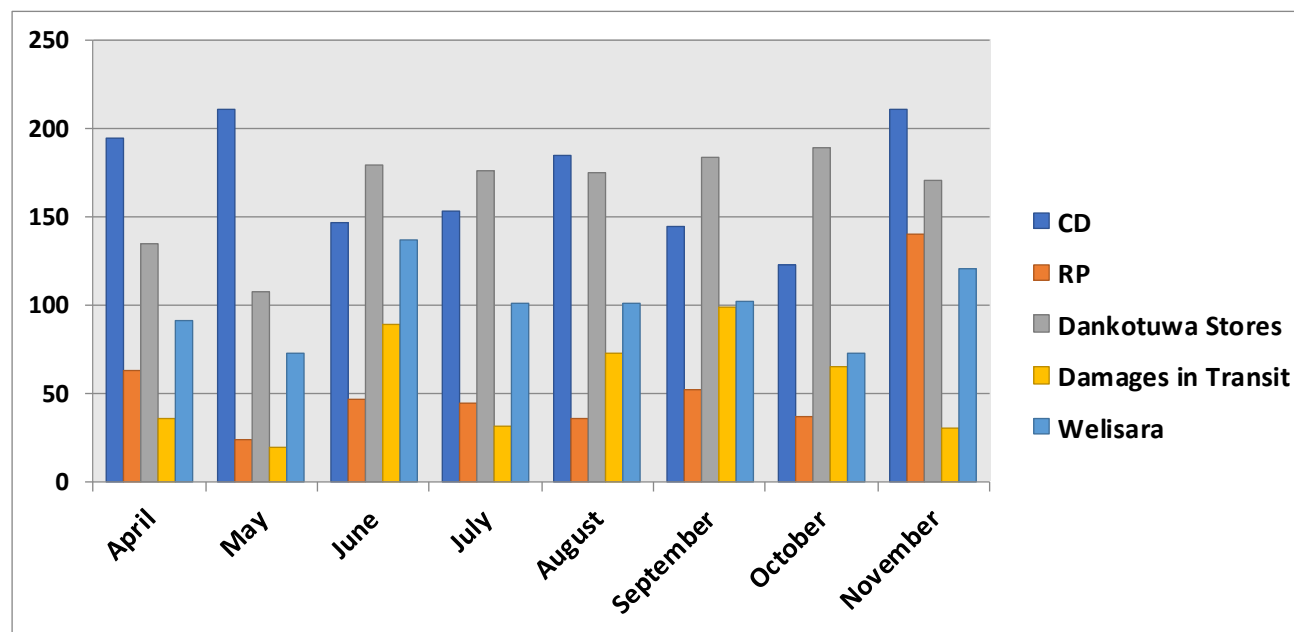


Out of **4360 damage pallets** between April to November;

-Highest damage pallets accounts from

CD PLANT= 1364 Pallets

-Second highest damage pallets accounts from **DANKOTUWA STORES= 1313 pallets**



Pallet Statistics in year 2019.



Damages

+



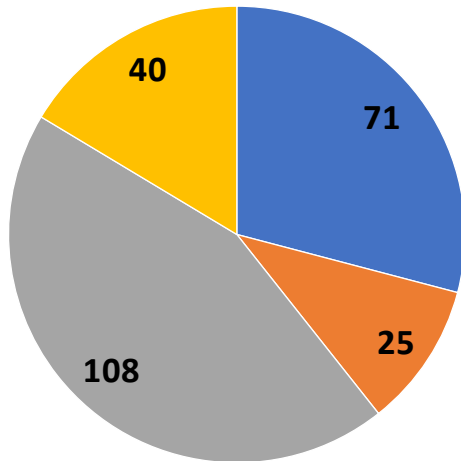
Disposals

+



Purchasing & Repair

Monthly Average = 243 Pallets

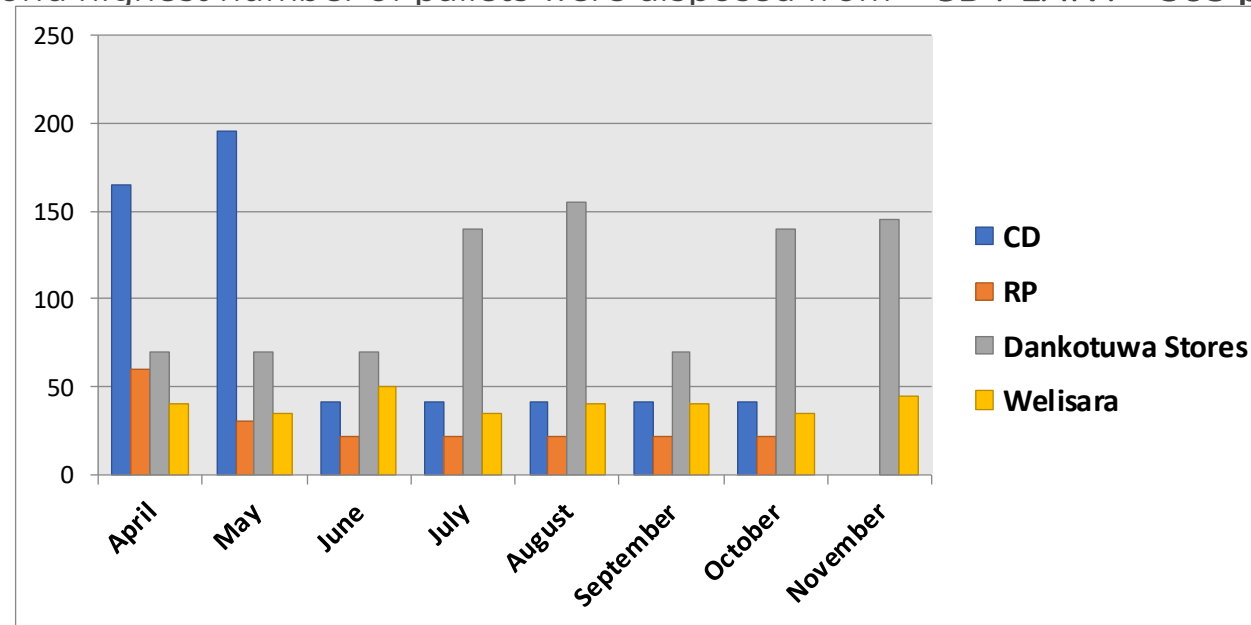


■ CD ■ RP ■ Dankotuwa Stores ■ Welisara

Out of 1945 disposed pallets between April to November;

-Highest number of pallets were disposed from **DANKOTUWA STORES= 860 pallets**

-Second highest number of pallets were disposed from **CD PLANT= 565 pallets**



Pallet Statistics in year 2019.



Damages

+



Disposals

+



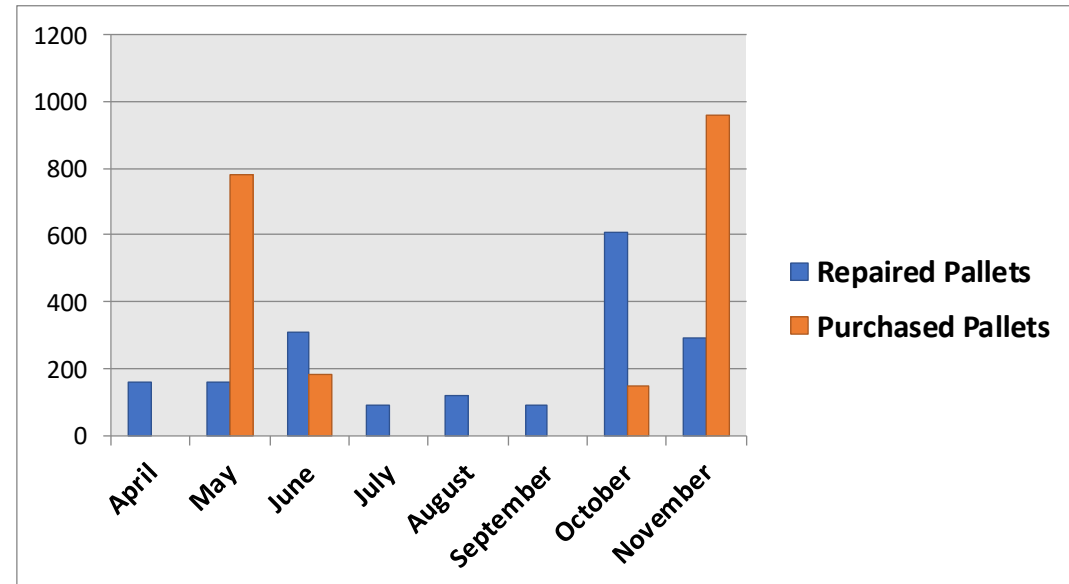
Purchasing & Repair



Total Purchasing Cost
1.3 Million Rupees

Total Repair Cost
0.12 Million Rupees

Total of 1830 Pallets have been repaired and 2072 Pallets have been purchased between April to November:



Review of Existing Process.

During the 2018/2019 financial year, there were instances with shortage pallets in the system.

Plant Wise Maximum Daily Capacity/Requirement			
	Capacity	Operational	Total
Material Stores	2500	70	2570
In Transit			215
PC Plant	125	35	160
Soap	120	65	185
Napkin	80	25	105
RPI Plant (Diva)	275	60	335
Huper/ CD (Diva)	1326	140	1466
PRI - Nappy Wash	60	10	70
Maneesha	60		60
FGWH	4250	130	4380
Total			9546

Operational Bottlenecks

1. Ownership is within the Dankotuwa warehouse and supported by Welisara warehouse for pallet logistics practices.
2. There isn't any proper mechanism to identify the pallet requirement before the lead time and not having the enough pallet suppliers to purchase within the lead time.
3. There isn't any further mechanism to check whether other plants are using the allocated pallets in the maximum way.
4. Issues related to loading, unloading and storing are not monitored due to volatility of the pallet requirements.

Proposed METHODOLOGY



PHASE 1



Statistics Inventory Model for RM, PM & FG

1. Requirement for standard norms.
2. Actual pallet consumption.
3. Maximum pallet allocation.
4. Implementation.

100% completed

PHASE 2



Identify the Existing Gaps in the Inventory Model

1. Compare the output of statistical inventory model.
2. Challenges for transformation.
3. Improvement opportunities.
4. Suggestions & Solutions.

100% completed

PHASE 3



Standardization model

1. Plan for breakage compensation.
2. Identify the triggering point.
3. Improvement opportunities.
4. Recommendations.

60% completed



Theoretical Part

Implementation

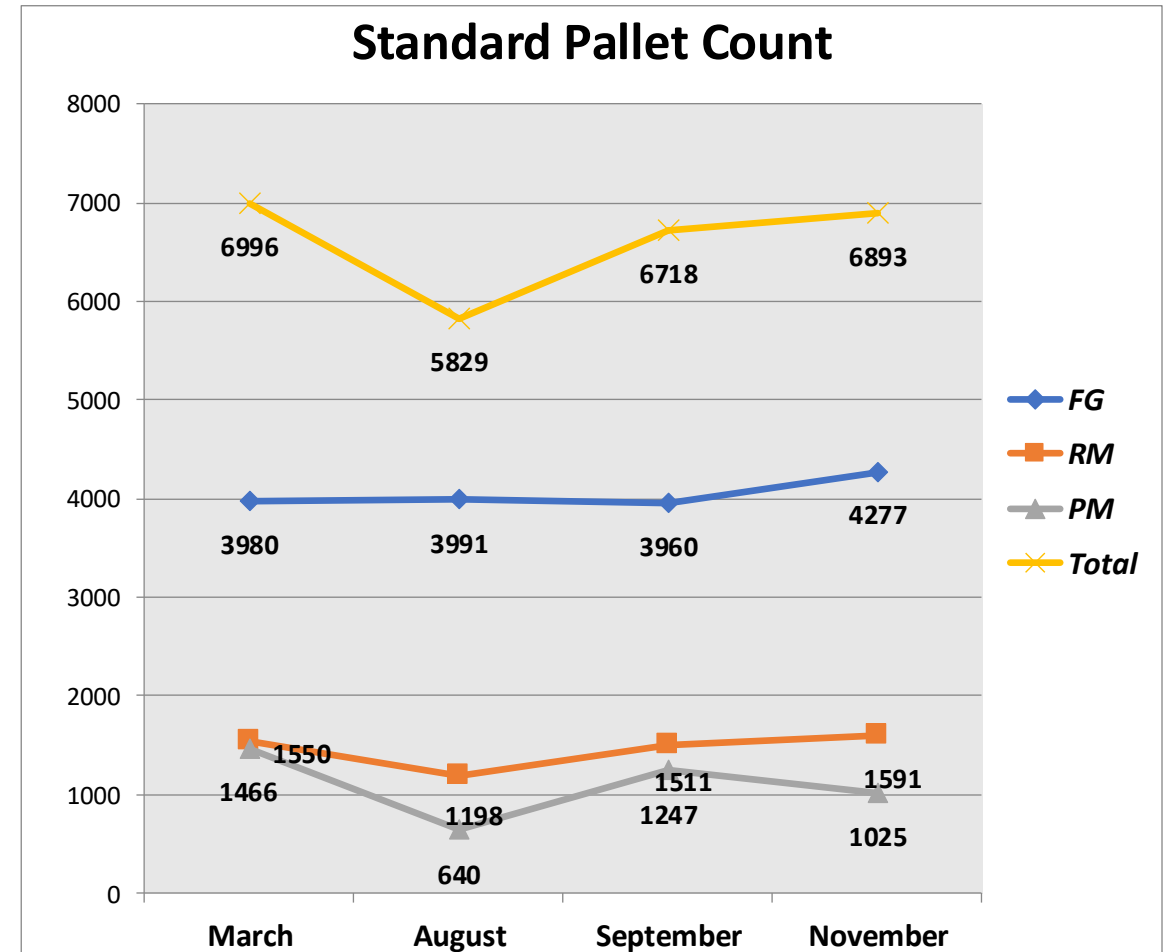
1. Statistics Inventory Model for RM, PM & FG

1.1 Requirement for Standard Norms

Model Assumptions:

1. Carried out the work for Supply Plan and Demand Plan for each month in year 2018.
2. Lead time is constant (Variation of lead time is zero)
3. Demand is randomly distributed
4. Storage is planned for maximum stock situations
5. Confidence level : 95%
6. Minimum Inventory Level (Buffer) = Hemas Standard Norm (RM-30 days, PM- 15 days).
7. Half pallet %- FG (15%), RM (15%), PM (30%)

	March	August	September	November
Storing	6996	5829	6718	6893
Operational Requirement	715	715	715	715
Total Pallets	7711	6544	7433	7608



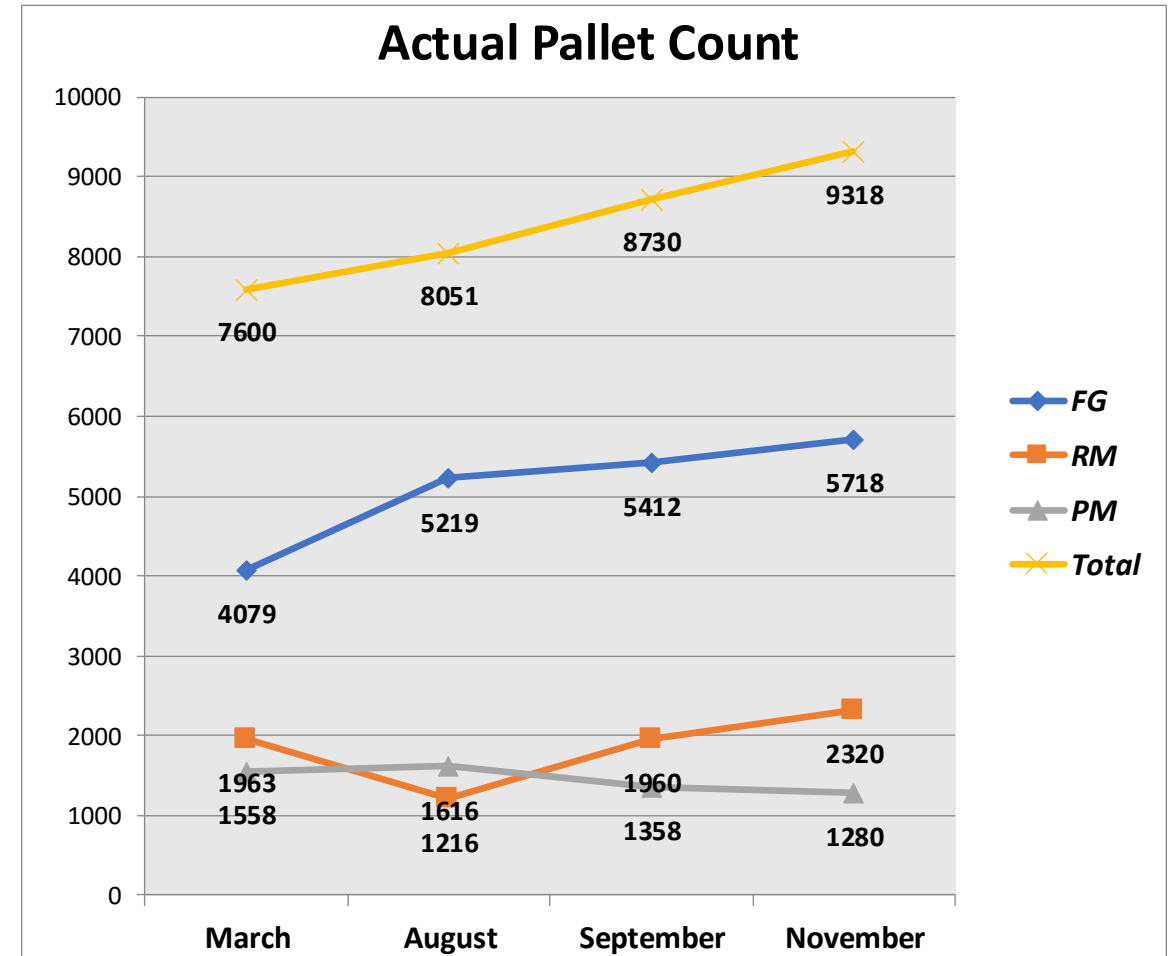
1.Statistics Inventory Model for RM, PM & FG

1.2 Actual Pallet Consumption

Model Assumptions:

1. Carried out the work for 15th day closing stock in each month.
2. Storage is planned for maximum stock situations
3. Half pallet %- FG (15%), RM (15%), PM (30%)

	March	August	September	November
Storing	7600	8051	8730	9318
Operational Requirement	715	715	715	715
Total Pallets	8315	8766	9445	10033



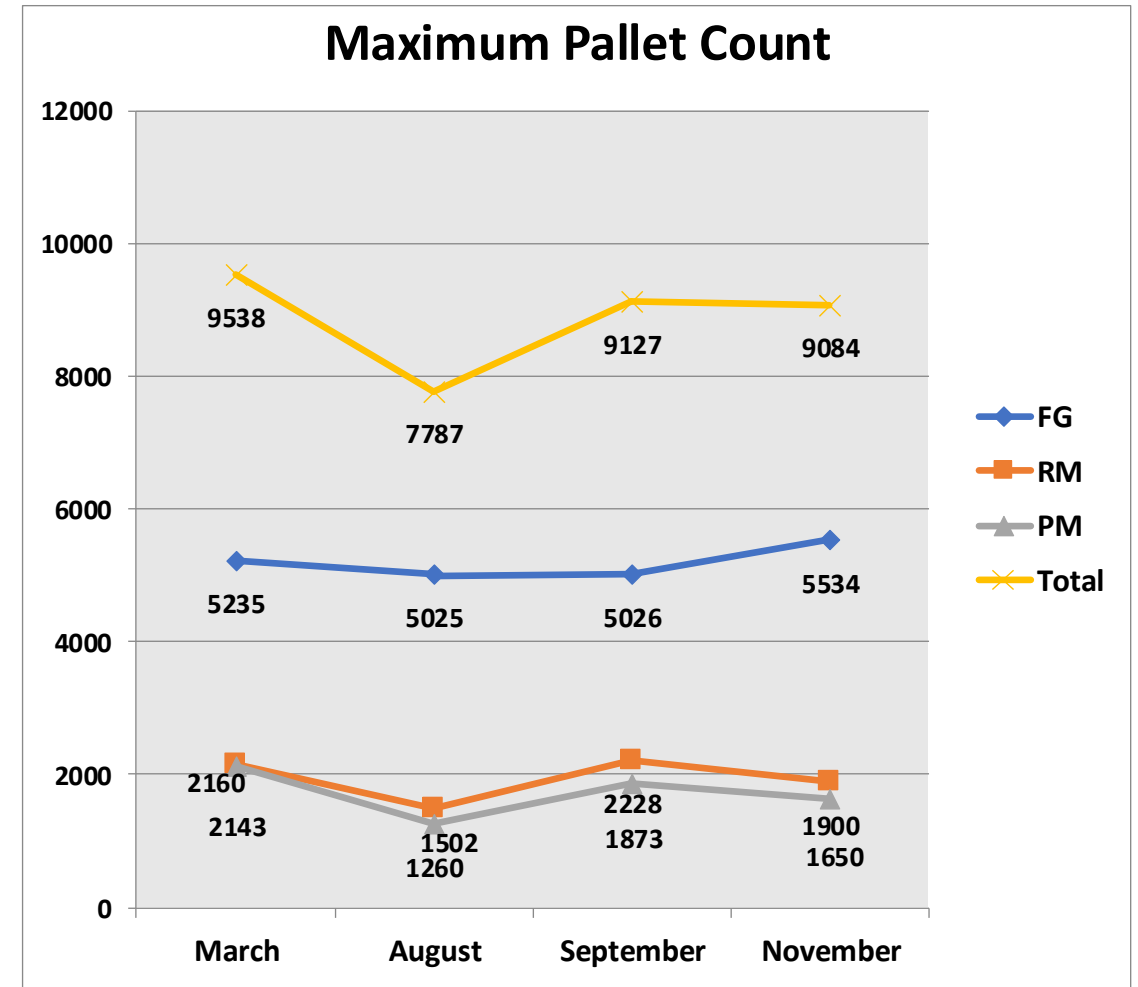
1.Statistics Inventory Model for RM, PM & FG

1.3 Maximum Pallet Requirement

Model Assumptions:

1. Carried out the work for Supply Plan and Demand Plan for each month in year 2018.
2. Lead time is constant (Variation of lead time is zero)
3. Demand is randomly distributed
4. Storage is planned for maximum stock situations
5. Confidence level : 95%
6. Maximum Inventory Level = Safety stock+ 2 *Cycle
7. Half pallet %- FG (15%), RM (15%), PM (30%)

	March	August	September	November
Storing	9538	7787	9127	9084
Operational Requirement	715	715	715	715
Total Pallets	10253	8502	9842	9799



1.Statistics Inventory Model for RM, PM & FG

1.4 Implementation

Model Assumptions:

1. Carried out the work for Supply Plan and Demand Plan for each month in year 2018.
2. Lead time is constant (Variation of lead time is zero)
3. Demand is randomly distributed
4. Storage is planned for maximum stock situations
5. Confidence level : 95%
6. Minimum Inventory Level= Standard Norms
7. Maximum Inventory Level = Safety stock+ 2*Cycle
8. Half pallet %- FG (15%), RM (15%), PM (30%)

	Actual
Storing	8505
Operational Requirement	715
Total Pallets	9220

Shortage Pallets as at 02.12.2019 = **223 Pallets**

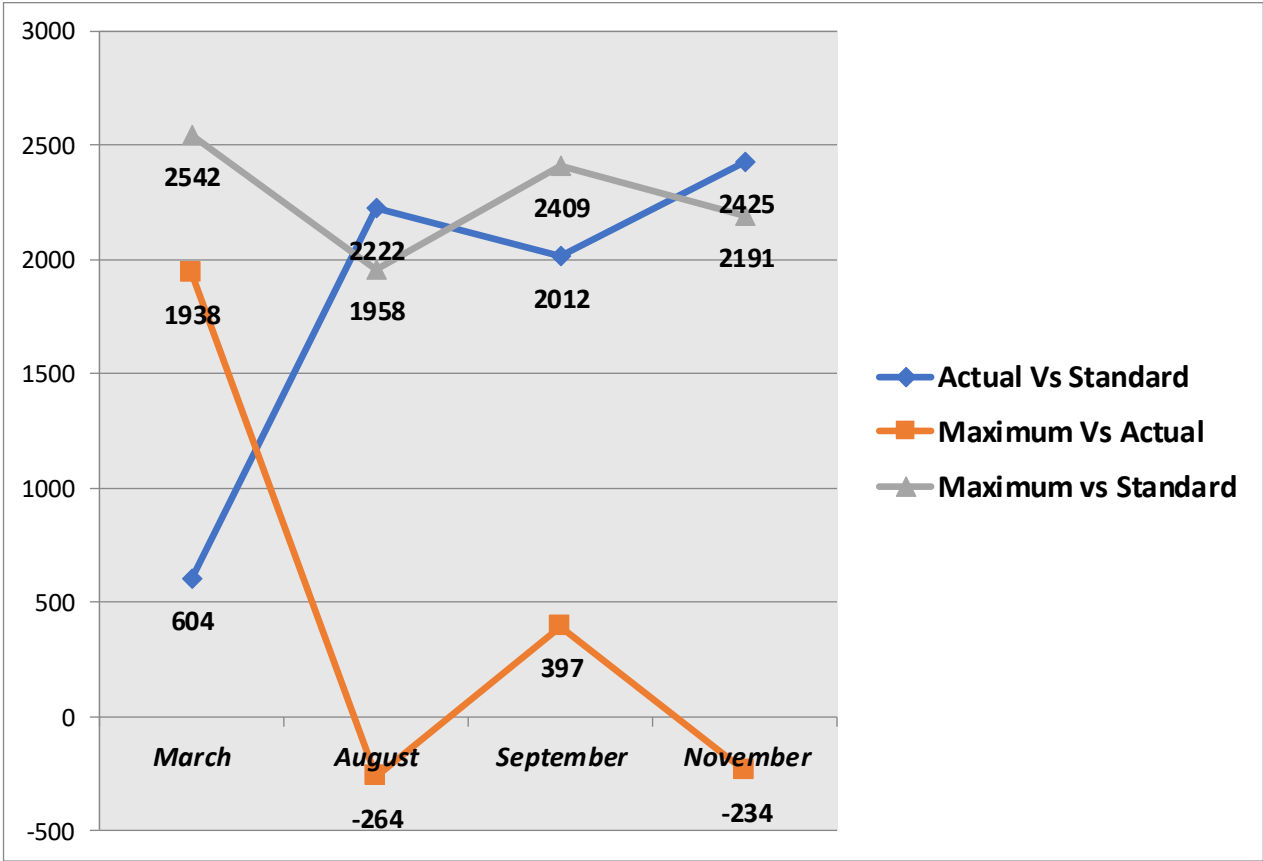
Pallet count 02.12.2019					
	FG	Material	Usable Empty	Empty to repair	Total
Material Stores	95	2428	353	143	3019
In Transit	47				47
Repairing Location				65	65
PC Plant	54	78	2	3	137
Soap	94	27	24	11	156
Napkin	6	48	6		60
RPI Plant (Diva)	102	184	151	15	452
Huper (Diva)	120	350	139	30	639
PRI - Nappy Wash	8	68	10	10	96
Value Added Service		485			485
FGWH	3706		95	40	3841
Total					8997

2. Identify the Existing Gaps in the Inventory Model

Revealing Existing Gaps in the System (Summary)

- 1. **Actual Vs Standard**- Actual pallet consumption rate is significant compares to standard norms.
- 2. **Maximum Vs Actual**- 10250 pallets (pool) are required to run a smooth operational flow. To maintain as the buffer.
- 3. **Maximum Vs Standard**- Significant difference is existing between maximum and the standard norm.
- 4. Inventory norms should be closely monitored to reduce the risk involved in lead time demand.

	Standard	Actual	Maximum
Average Count	7324	9140	9760
Maximum Scenario	7711	10033	10253
Benchmark (90% of the Maximum) = 9420 Pallets	800 Pallets should be Outsourced in the month of November		



3. Standardization Model

Improvement Opportunity;



Providing ownership for damage pallets

- Can identify the requirement shortfall with the buffer?
- Address the purchasing lead time + repair lead time (Minimum of 200 pallets additionally).



Strengthen the pallets and seek for standard pallets from the direct suppliers.

- Are there are any deliveries with standard pallets that can accommodate in racking systems?
- Seek for potential pallet suppliers with high durability and strength.

Challenges for Transformation;

Determining the triggering point

- What is the future pallet purchasing pattern?
- Who is responsible for the pallet breakage?
- Can we overlook the process through pallet reconciliation?
- How to address to peak requirements?

Convincing suppliers to get the standard pallets & determining the potential pallet suppliers that can align to our specifications?

- Might have to incur high cost to accommodate strengthen pallets.

Recommendation;

- **Plant manager** should take the responsibility and raise direct PR to purchasing team.
- Initiate a **colour coding system** for **freeze pallets** (non movable) in the plant.
- **Movable:** Monthly reconciliation + monthly breakage % should be communicated directly.
- Lead time pallet requirement should be communicated by the planning and individual plant to the pallet procurement team.
- Select an **one cut off point** and initiate the pallet reconciliation.
- Seek for **hiring opportunity** to address kick off and peak demands.

THANK YOU FOR LISTENING